Introduction

Project(Capstone Project)

Q] Star schema Vs SnowFlake Schema

Star Schema: Dimension tables are denormalized, making queries simpler and faster.

Snowflake Schema: Dimension tables are normalized, reducing data duplication.

Q] RightTrim and Left Trim in SQL (removing special character)

LTRIM(): Removes spaces or specified characters from the left side of a string.

RTRIM(): Removes spaces or specified characters from the right side of a string.

SELECT LTRIM('###Hello', '#');

-- Output: Hello

SELECT RTRIM('Hello###', '#');

-- Output: Hello

Q] What Trim Function will do?

The `TRIM()` function in SQL removes unwanted spaces or specified characters from both the left and right sides of a string.

Data Warehousing

Data warehousing is the process of collecting and storing data from multiple sources in one central location for reporting and analysis.

Remember: Data Warehouse = Centralized data + Analysis + Decision-making.

If we have more joins does the performance will be degraded?

Yes, more joins can degrade query performance because SQL needs to combine data from multiple tables, but proper indexing and optimized queries can improve performance.

Types of SQL joins

INNER JOIN: Returns only matching records from both tables.

LEFT JOIN: Returns all records from the left table and matching records from the right table.

RIGHT JOIN: Returns all records from the right table and matching records from the left table.

FULL OUTER JOIN: Returns all records from both tables, including matching and non-matching records.

CROSS JOIN: Returns every possible combination of rows from both tables.

SELF JOIN: Joins a table with itself to compare related records.

Python is structured or object oriented language?

Python is both a structured (procedural) and object-oriented programming language because it supports functions, step-by-step programming, and classes and objects.

Types of operators

Arithmetic: +, −, *, /

Comparison: ==, !=, >, <

Logical: `and`, `or`, `not`

Assignment: `=`, `+=`, `-=`

Bitwise: `&`, `|`, `^`

Membership: `in`, `not in`

Identity: `is`, `is not`

What's is() Operator?

The `is` operator checks whether two variables refer to the same object

How to Remove duplicates from list? [With Code or Syntax]

We can remove duplicates from a Python list using the `set()` function.

```
my_list = [1, 2, 2, 3, 4, 4, 5]
unique_list = list(set(my_list))
print(unique_list)
```

Fact Table and Dimension table

Fact Table: Stores quantitative data (measurable values), such as sales amount, quantity, and profit.

Dimension Table: Stores qualitative or descriptive data, such as customer name, product name, and location.

What types of Keys are stored in Fact Table

Answer: A Fact Table mainly contains foreign keys that connect to dimension tables and measurements such as sales amount and quantity. It may also contain a primary or composite key, depending on the design.

What is Granularity?

Granularity means how detailed the data is stored in a fact table.

Example:

- **High granularity (detailed): One row for every individual sale.Low granularity (summarized): One row for total sales per day.**
- **Interview answer: Granularity defines the level of detail stored in a fact table. The finer the granularity, the more detailed the data.**

In Sql, How will you Identify duplicate records (SQL Query)?

```
SELECT Name, COUNT(*)
FROM Students
GROUP BY Name
HAVING COUNT(*) > 1;
```

To identify unique records?

```
SELECT Name, COUNT(*)
FROM Students
GROUP BY Name
HAVING COUNT(*) = 1;
```

If trying to insert duplicate in unique primary key?

If we try to insert a duplicate value into a primary key column, SQL will generate a primary key violation error because primary key values must be unique and cannot be NULL.

Can primary key be NULL?

No, a primary key cannot contain NULL values because it uniquely identifies each record in a table.

What is Delete,Drop & Truncate?

1. DELETE: Removes specific rows from a table using the WHERE condition, or all rows if no condition is given.

2. DROP: Removes the entire table structure along with its data.
3. TRUNCATE: Removes all rows from a table but keeps the table structure.

If we write Delete without any Condition , will it act similar to Truncate?

No, both remove all rows, but they work differently. DELETE Can be rolled back if executed inside a transaction that has not been committed. But Truncate cannot

Difference between PowerBi, Tableau,QlikSense? (WHICH IS GOOD FOR ANALYSIS?)

"Power BI is known for ease of use and Microsoft integration, Tableau for powerful visualizations, and Qlik Sense for associative data exploration. The best tool depends on the business requirements."

What is SCD? [With Types]

SCD stands for Slowly Changing Dimension. It is used in data warehousing to manage changes in dimension data over time, such as a customer's address.

Types of SCD:

Type 0: Keeps the original data unchanged.Type 1: Updates the existing data without maintaining history.Type 2: Creates a new record to maintain complete history.Type 3: Adds a new column to store limited history, such as the previous address.

Remember for your interview:

Type 0 — No changes.Type 1 — Overwrite.Type 2 — New row.Type 3 — New column.

What is palette in python?

Palette = A predefined collection of colors used in data visualization.

What is tuple?

Tuple = Ordered + Immutable, and it uses round brackets `()`.

Break,Continue and Pass?

1. `break`: Terminates the loop immediately.

2. `continue`: Skips the current iteration and moves to the next iteration.

3. `pass`: Does nothing; it is used as a placeholder for future code.

`break` = Stop, `continue` = Skip, `pass` = Do nothing.

What is Surrogate Key?

Remember: Surrogate key = Artificially generated unique ID.

What is Confirm Dimension And Role-Playing Dimension?

Conformed Dimension = Shared across multiple fact tables.

Role-Playing Dimension = Same dimension used for different roles.

Window And Date Related Functions in Sql

Window functions perform calculations across a set of rows without combining them into a single row.

Explain Data types of Python?

`int`, `float`, `str`, `bool`, `list`, `tuple`, `set`, and `dict`

What is Type Casting?

Type casting = Converting one data type into another using functions like `int()`, `float()`, and `str()`.

Difference between Slicer and Filter in PowerBi? How many types of Filter?

Slicer: An interactive visual used to filter data on a report.

Filter: Restricts data at the visual, page, or report level.

Types:

Visual-level filter: Filters data for a single visual.

Page-level filter: Filters all visuals on a particular page.

Report-level filter: Filters all pages in the report.

Drillthrough filter: Passes selected values to another report page to show detailed information.

How to do you provide interactivity in PowerBi?

We provide interactivity in Power BI using slicers, filters, drill-through, tooltips, and visual interactions between charts.

What is Row-Level Security?

Row-Level Security (RLS) in Power BI restricts data access for users based on their roles, so each user can see only the rows they are authorized to view.

How to Publish report in PowerBi?

To publish a report in Power BI, click Publish in Power BI Desktop, select the destination workspace, and click Select. The report will be published to the Power BI Service.

What is Parameter?

A parameter in Tableau is a dynamic variable that allows users to change a value to control calculations, filters, or visualizations in a dashboard.

How do you Publish the Tableau Dashboard to the server?

To publish a Tableau dashboard, go to Server → Publish Workbook, sign in to Tableau Server or Tableau Cloud, select the project, and click Publish.

What is difference between join and Blend?

Join: Combines tables at the data source level.

Blending: Combines data from different data sources at the visualization level.

Include expressions in Tableau Calculated field?

In Tableau, the `INCLUDE` expression is a Level of Detail (LOD) expression that adds a specified dimension to the level of detail of a calculation.

Syntax: `{ INCLUDE [Dimension] : AGG([Measure]) }`

Example: `{ INCLUDE [Customer Name] : SUM([Sales]) }`

How to add Tooltip?

In Tableau, select the worksheet, drag the required fields to the Tooltip shelf on the Marks card, then click Tooltip to edit the text displayed when hovering over a data point.

What are different Filters in Tableau?

Extract Filter – Filters data while creating an extract.

Data Source Filter – Filters data at the data source level.

Context Filter – Creates a primary filter that other filters depend on.

Dimension Filter – Filters categorical data, such as names or categories.

Measure Filter – Filters numerical data, such as sales or profit.

Table Calculation Filter – Filters results based on table calculations.

Any customized visual we have used in Tableau or PowerBi?

Yes, in Power BI, we can use custom visuals such as Sankey charts, Word Clouds, and advanced maps from AppSource. In Tableau, we

can create customized visuals using calculated fields, dual-axis charts, and different chart types.

Expression to concatenate two names in Tableau?

In Tableau, we use the + operator to concatenate two names.

[First Name] + " " + [Last Name]

What is workspace in PowerBi?

A workspace in Power BI is a collaborative area where users create, store, manage, and share reports, dashboards, and semantic models.

What is decoraters in Python

Decorator = Adds functionality to an existing function.(used to modify or extend the behavior of a function)

Iterators in Python?

An iterator in Python is an object that allows us to access elements one by one using `__iter__()` and `__next__()` methods.

What is difference between join and merge?

In Python, `merge()` and `join()` are used to combine DataFrames in Pandas. `merge()` combines DataFrames using common columns or keys, while `join()` combines them mainly using their indexes.

What are nested queries?

A nested query, also called a subquery, is a SQL query written inside another query. The inner query provides results that are used by the outer query.

What are Variables in python?

Variables in Python are names used to store data values in memory.

Few input/output Operations in Python?

`input()` = Takes input, `print()` = Displays output.

What are conditional statements?

Conditional statements in Python are used to execute different blocks of code based on whether a condition is True or False. The main types are `if`, `if-else`, `if-elif-else`, and nested `if`.

Loops in python?

Loops in Python are used to execute a block of code repeatedly. Python has two main types of loops: `for` loop, used to iterate over a sequence, and `while` loop, used to repeat code as long as a condition is true.

Lambda function in python?

A lambda function in Python is a small anonymous function defined using the `lambda` keyword. It can take multiple arguments but contains only one expression.

Navigations ?

Power BI: Navigation allows users to move between report pages using buttons, page navigators, bookmarks, and page tabs.

Building blocks in Qlik?

The main building blocks of Qlik Sense are Dimensions, Measures, and Visualizations. Dimensions categorize data, measures perform calculations, and visualizations present data in charts, tables, and graphs.

Trim query in sql to show first name and cut down the surname

```
SELECT
  CustomerName,
  LEFT(CustomerName, CHARINDEX(' ', CustomerName) - 1) AS FirstName
FROM Customers;
```

How to refresh data? Extract refresh And Datasource refresh?

(Tableau): We can refresh data by clicking Data → [Data Source] → Refresh for a live connection, or Data → [Data Source] → Extract → Refresh for an extract.
Extract refresh updates the stored extract; data source refresh updates or reconnects the data source.

How to Schedule Data Refresh?

To schedule a data refresh, publish the workbook or data source to Tableau Server or Tableau Cloud, go to Data Sources → Extract Refreshes, select New Extract Refresh, and set the refresh schedule.